PUNJAB ENGINEERING COLLEGE (DEEMED TO BE UNIVERSITY)
Sector 12, Chandigarh - 160012 | Office of Dean Academic Affairs (DAA)
Capstone Coursework Deliverable • Academic Clearance Automation Engine

Software Requirements Specification (SRS)

Academic Grievance, No-Dues & Clearance Automation Engine

Institution: Punjab Engineering College (Deemed to be University), Sector 12, Chandigarh - 160012
Document Identifier: PEC-SE-SRS-2026-FROZEN
Standard Compliance: IEEE Std 830-1998 (Recommended Practice for Software Requirements Specifications)
Baseline Version: 3.0.0 (FROZEN ARCHITECTURAL BASELINE)
Administrative Authority: Office of the Dean Academic Affairs (DAA), PEC Chandigarh
Classification: Institutional Standard Specification
Status: FROZEN & RATIFIED (Implementation-Agnostic Baseline)

Baseline Freeze Declaration & Change Control Policy

> **NOTICE OF BASELINE FREEZE:**
> This specification defines the immutable architectural, functional, data, and behavioral contracts for the PEC Academic Grievance, No-Dues & Clearance Automation Engine. All functional boundaries, mathematical formulas, state transitions, security constraints, and data definitions in this document are **FROZEN**.
>
> Any future implementation-level modifications—including UI component updates, styling alterations, backend database driver migrations (e.g., SQLite to PostgreSQL), or API transport optimizations—are explicitly encapsulated within the interface abstractions defined herein and **shall not invalidate or alter this SRS, the system DFDs, or the UML structural and behavioral models**.

1. Introduction

1.1 Purpose

This document establishes the official and complete software requirements specification for the **Academic Grievance, No-Dues & Clearance Automation Engine** deployed at **Punjab Engineering College (Deemed to be University), Sector 12, Chandigarh**.

The system replaces manual paper-based clearance circulars with an automated, workflow-driven, cryptographically secured clearance engine governing student clearance across central campus facilities, departmental academic laboratories, and the finance branch.

1.2 Scope of the System

The system encompasses the following functional domains:

1. **Identity & Positional Authority Resolution:** Authentication of PEC students via immutable 8-digit Student Identification Numbers (SIDs, e.g., `22103042`) and department officials via permanent administrative designations (e.g., *Dean Academic Affairs*, *Chief Warden Hostels*, *Librarian*, *President Gymkhana*, *HOD Computing & Labs*, *Accounts Officer*).
2. **Multi-Gate Clearance Workflow Engine:** Enforcement of a 3-tier clearance pipeline using the Chain of Responsibility design pattern:
   - **Gate 1 (Concurrent / Parallel):** Dr. B.R. Ambedkar Central Library, PEC Hostels (Shivalik, Kurukshetra, Himalaya, Aravali, Vindhya, Kalpana Chawla), and Gymkhana & Sports Complex.
   - **Gate 2 (Sequential / Prerequisite-Enforced):** Departmental Computing and Research Laboratories (CSE, ECE, Mechanical, Electrical, Civil, Aerospace, Metallurgy, Production).
   - **Gate 3 (Terminal Settlement & Refund Release):** Accounts & Finance Branch for tuition reconciliation and refundable Caution Money release.
3. **Cryptographic Integrity & Digital Seals:** Departmental approvals sealed with Keyed-Hash Message Authentication Codes ($\mathrm{HMAC\text{-}SHA256}$); issued certificates sealed with cryptographic digest hashes ($\mathrm{SHA\text{-}256}$).
4. **SLA-Driven Grievance Arbitration:** Dynamic service level agreement calculation with priority queues and automated escalation to the Dean Academic Affairs upon deadline breach.
5. **Public Verification & Tamper Evidence:** Public, unauthenticated verification gateway for employers, higher education institutions, and visa authorities to validate certificate authenticity.

1.3 Definitions, Acronyms, and Abbreviations

- **PEC:** Punjab Engineering College (Deemed to be University), Chandigarh.
- **SID:** Student Identification Number (8-digit institutional alphanumeric code, format: `YYBBBRRR`, e.g., `22103042`).
- **DAA:** Dean Academic Affairs, Punjab Engineering College.
- **DSA:** Dean Student Affairs, Punjab Engineering College.
- **Central Library:** Dr. B.R. Ambedkar Central Library, PEC Sector 12 campus.
- **Hostel Council:** Administrative body governing PEC residences (Shivalik, Kurukshetra, Himalaya, Aravali, Vindhya, KCH).
- **Gymkhana:** PEC Sports Complex, Gymnasium, Grounds, and Swimming Pool facility.
- **Caution Money:** Refundable institutional deposit collected upon admission, reconciled and released at Gate 3.
- **CoR:** Chain of Responsibility Behavioral Design Pattern.
- **SLA:** Service Level Agreement governing grievance dispute resolution timelines.
- **HMAC:** Keyed-Hash Message Authentication Code (RFC 2104).
- **INR / ₹:** Indian Rupee, the sole monetary currency of the institution.

1.4 References

- IEEE Std 830-1998: *IEEE Recommended Practice for Software Requirements Specifications*.
- FIPS PUB 180-4: *Secure Hash Standard (SHS)*, National Institute of Standards and Technology.
- RFC 2104: *HMAC: Keyed-Hashing for Message Authentication*.

- PEC Academic Ordinances & Regulations for B.Tech., M.Tech., and Ph.D. Degree Programs (2022-2026).
- UGC / AICTE Guidelines on Transparent Digital Grievance Redressal in Higher Educational Institutions.

1.5 IEEE Std 830-1998 Quality & Compliance Matrix

This document conforms strictly to the **IEEE Std 830-1998** standard requirements for well-formed software specifications:

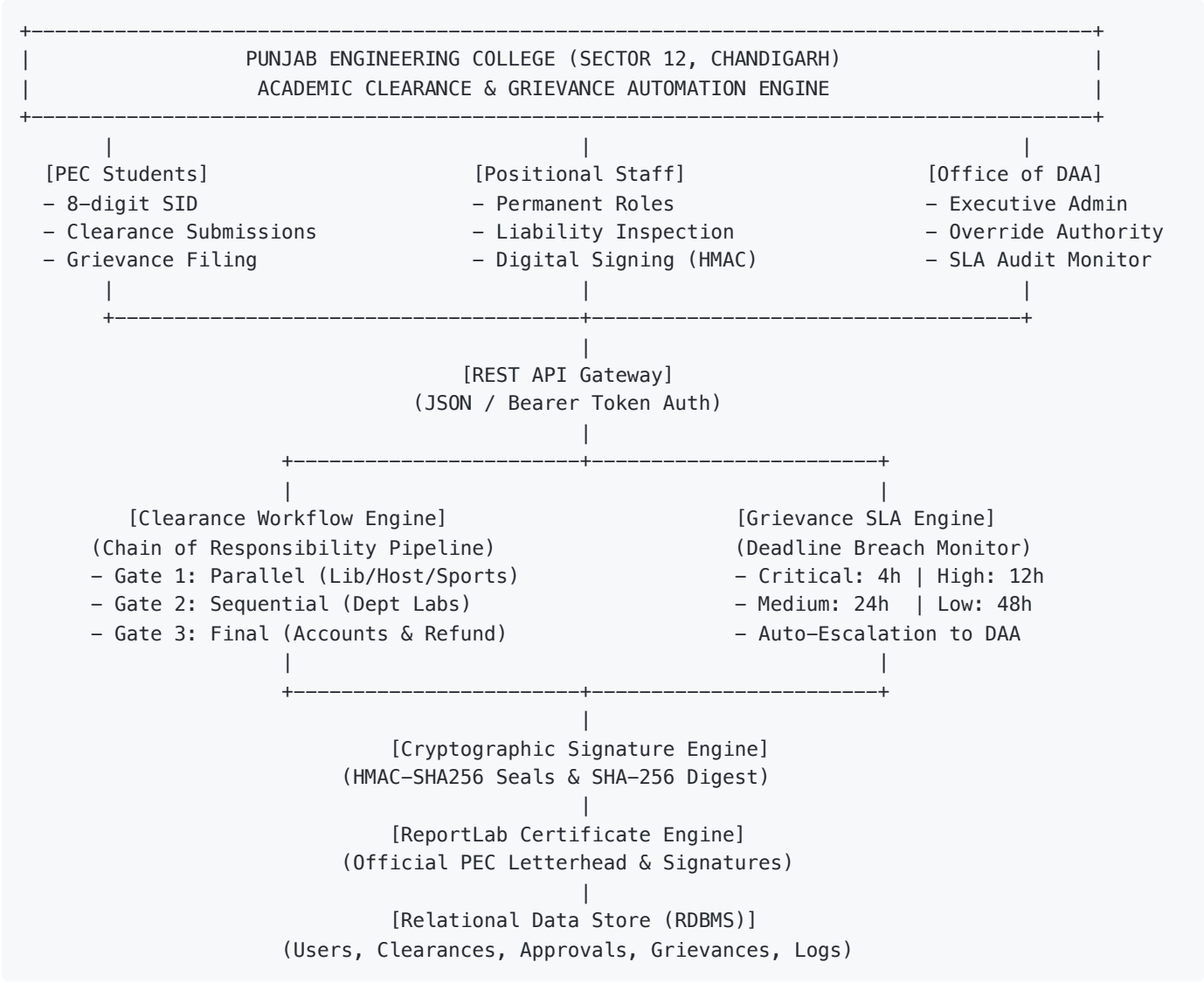
IEEE 830 Quality Attribute	Specification Fulfillment	Section Reference
1. Correctness	Every stated requirement corresponds directly to institutional regulations of Punjab Engineering College (Sector 12, Chandigarh).	Sections 1.2, 3.2
2. Unambiguity	Every functional requirement possesses an explicit, unique identifier (`FR-X.Y`) with quantifiable inputs, outputs, and mathematical formulas.	Section 3.2
3. Completeness	Covers all institutional clearance departments, fee reconciliation, SLA grievance tracking, and cryptographic certificate minting.	Sections 2.1, 3.2
4. Consistency	Gate progression constraints, actor permissions, and state transitions are mathematically non-contradictory across all modules.	Sections 3.2, 5
5. Ranked for Importance / Stability	Core 3-gate pipeline and cryptographic seals are designated as **Essential & Frozen Invariants**; analytics and export tools are supportive.	Section 5
6. Verifiability	Every functional and non-functional requirement is objectively testable via automated unit tests, static analysis, or end-to-end API calls.	Section 4, Test Suite
7. Modifiability	Organized hierarchically into decoupled modules, allowing implementation adjustments without altering requirements contracts.	Sections 3.1 - 3.2
8. Traceability	Forward and backward traceability maintained from user personas $\rightarrow$ functional requirements $\rightarrow$ DFD processes $\rightarrow$ UML classes.	Sections 2.2, 3.2

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2. Overall Description

2.1 Product Perspective & Context

The system serves as the institutional clearance and no-dues operating core. It integrates student records, departmental liability registries, and financial reconciliation ledgers into an event-driven workflow engine:



2.2 User Personas & Characteristics

1. **Student Persona:**
   - Possesses a valid 8-digit PEC SID (`22103042`) and `@pec.edu.in` credentials.
   - Capable of creating a single active clearance request per academic milestone.
   - Views live gate status, outstanding departmental liabilities in INR (`₹`), and signs off on fee settlements.
   - May file priority-ranked grievances contesting erroneous dues.
2. **Department Positional In-Charge Persona:**
   - Permanent administrative designation (not tied to transient personal names).
   - Responsible for verifying physical asset returns and financial dues within their department.
   - Executes cryptographic digital signing upon clearance approval.
   - Possesses authority to record outstanding dues in INR (`₹`) or resolve contested grievances.
3. **Dean Academic Affairs (DAA / Admin Persona):**
   - Highest academic administrative authority at PEC Chandigarh.
   - Possesses read-write visibility across all departments and students.
   - Authorized to execute administrative overrides (`ADMIN_OVERRIDE`) to unblock disputed approvals with mandatory audit logging.

    - Monitors SLA escalation dashboards and institutional performance analytics.
4. **Public Verifier Persona:**
    - External entities (employers, university admissions offices, verification agencies).
    - Requires zero login credentials.
    - Validates official certificates using the immutable SHA-256 digital certificate hash.

2.3 General Constraints

1. **Institution-Specific Data Constraints:**
    - Student IDs strictly follow the 8-digit PEC format: `YYBBBRRR` (e.g., `22` = Batch 2022, `103` = Computer Science, `042` = Roll index).
    - Institutional currency is strictly Indian Rupees (`₹` / INR).
    - Staff accounts represent institutional designations, ensuring multi-year stability regardless of faculty rotation.
2. **Regulatory & Architectural Constraints:**
    - Gate 2 (Department Labs) cannot unlock until all Gate 1 departments are `APPROVED`.
    - Gate 3 (Accounts) cannot unlock until Gates 1 and 2 are fully `APPROVED`.
    - A clearance certificate can only be minted if all 5 constituent department gates are `APPROVED`.
    - Every approval must contain an HMAC-SHA256 digital signature seal generated with an institutional server secret.

3. Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

- **Student Dashboard:** Responsive Single Page Application (SPA) displaying live gate progression across Gate 1 (Library, Hostels, Sports), Gate 2 (Labs), and Gate 3 (Accounts), itemized liabilities in INR (`₹`), and grievance filing modals.
- **Department Staff Console:** Department-filtered review queue enabling liability verification, dues input in INR (`₹`), resolution notes, and one-click cryptographic HMAC signing.
- **Dean Academic Affairs (DAA) Portal:** Executive analytics dashboard displaying institutional bottleneck metrics, overdue SLA breach alerts, student rosters, and administrative override controls.
- **Public Certificate Verification Interface:** Tokenless, unauthenticated web form accepting a certificate ID or SHA-256 hash and rendering instant verification credentials.

3.1.2 Hardware Interfaces

- **Server Tier:** Architecture-neutral x86-64 / ARM64 execution environment (minimum 1 vCPU, 512 MB RAM).
- **Client Tier:** Standard computing workstation or mobile browser supporting ECMAScript 2022+ and HTML5 Canvas.

3.1.3 Software Interfaces

- **Operating System:** Platform independent (macOS, Linux / POSIX, Windows).

- **Application Server Gateway:** ASGI-compliant HTTP server (Uvicorn / FastAPI).
- **Database Engine Interface:** SQL-standard Relational Database Management System (SQLite3 embedded / PostgreSQL 14+) interfaced via SQLAlchemy Object-Relational Mapping (ORM).
- **Cryptographic Subsystem:** Cryptographic hashing libraries providing FIPS-compliant SHA-256 and HMAC-SHA256 primitives.
- **Document Rendering Subsystem:** ReportLab vector graphics engine for PDF generation.

3.1.4 Communications Interfaces

- **Network Protocol:** HTTP/1.1 and HTTP/2 over TLS 1.3 (HTTPS).
- **Data Exchange Format:** JavaScript Object Notation (JSON, RFC 8259) with UTF-8 character encoding.
- **Authorization Header:** HTTP `Authorization: Bearer <JWT_TOKEN>` standard header for stateless session validation.

3.2 Functional Requirements (System Features)

Module 1: Authentication & Identity Management

- **FR-1.1 [Institutional Credential Verification]:** The system shall authenticate users using secure credentials. Passwords shall be hashed using salted one-way hashing algorithms (Bcrypt).
- **FR-1.2 [Role & Scope Enforcement]:** The system shall enforce role-based access control (RBAC) across three principal scopes: `STUDENT`, `STAFF`, and `ADMIN`.
- **FR-1.3 [Positional Designation Binding]:** Staff profiles shall be bound to institutional department codes (`LIBRARY`, `HOSTEL`, `SPORTS`, `LABS`, `ACCOUNTS`) and positional titles rather than personal identities.

Module 2: Clearance Request Lifecycle

- **FR-2.1 [Single Active Clearance Constraint]:** A student may have at most one clearance request in `PENDING` status at any given time.
- **FR-2.2 [Milestone Justification Recording]:** The clearance request must record the academic milestone:
  - B.Tech Final Degree Graduation
  - Mandatory 6-Month Industrial Training / Internship (Semester 7/8)
  - M.Tech / Ph.D. Thesis Submission
  - Program Withdrawal / Caution Money Settlement
- **FR-2.3 [Automated Approval Gate Spawning]:** Upon creation of a clearance request, the system shall instantiate exactly five departmental clearance records:
  - `LIBRARY` (Gate 1, initialized to `PENDING`)
  - `HOSTEL` (Gate 1, initialized to `PENDING`)
  - `SPORTS` (Gate 1, initialized to `PENDING`)
  - `LABS` (Gate 2, initialized to `WAITING_PREREQUISITE`)
  - `ACCOUNTS` (Gate 3, initialized to `WAITING_PREREQUISITE`)

Module 3: Chain of Responsibility (CoR) Approval Gating

- **FR-3.1 [Gate 1 Concurrent Evaluation]:** The Library, Hostel, and Sports Complex handlers shall evaluate and update clearance statuses concurrently and independently.

- **FR-3.2 [Gate 2 Prerequisite Enforcement]:** The Departmental Labs handler shall reject approval attempts if any Gate 1 department status is not `APPROVED`, unless an explicit `ADMIN_OVERRIDE` has been executed by the DAA.
- **FR-3.3 [Gate 3 Financial & Caution Money Settlement]:** The Accounts handler shall reject approval attempts if Gate 1 or Gate 2 is incomplete. Upon approval, Accounts reconciles institutional fees and authorizes the release of the student's Caution Money deposit.
- **FR-3.4 [Digital Signature Generation]:** Upon approving any gate, the handler shall compute a keyed digital signature seal: \text{Seal} = \text{HMAC-SHA256}\Big(K_{\text{secret}}, ;\text{staff_id} \parallel \text{dept_code} \parallel \text{request_id} \parallel \text{timestamp}\Big)
- **FR-3.5 [Terminal Status Propagation]:** When Gate 3 reaches `APPROVED`, the master clearance request status shall transition to `APPROVED`, and the system shall automatically mint a unique Certificate ID (`PEC-CLR-YYYY-XXXXX`) and an immutable SHA-256 certificate hash.

Module 4: Administrative Authority & Concurrency Control

- **FR-4.1 [Executive Override Authority]:** The Dean Academic Affairs (Admin) shall have authority to override any departmental rejection, transitioning the gate to `APPROVED` with an audit flag `ADMIN_OVERRIDE`.
- **FR-4.2 [Atomic Gate Recalculation]:** All prerequisite checks and gate unlocking transitions shall execute within serialized database transactions to prevent race conditions during concurrent staff reviews.
- **FR-4.3 [Audit Trail Immutability]:** Every approval, rejection, due modification, grievance resolution, and override shall append an immutable record to the institutional audit log containing timestamp, actor ID, action type, and JSON metadata.

Module 5: Grievance Arbitration & SLA Escalation

- **FR-5.1 [Grievance Filing]:** A student may lodge a formal grievance against any department reporting outstanding dues, specifying subject, description, priority, and disputed amount in INR (`₹`).
- **FR-5.2 [Dynamic SLA Computation]:** The system shall compute an exact resolution deadline upon grievance registration based on priority: $$\text{Deadline} = \text{Timestamp}*{|text{submission}} + |Delta t*{\text{priority}}$$
  - `CRITICAL`: $\Delta t = 4$ hours
  - `HIGH`: $\Delta t = 12$ hours
  - `MEDIUM`: $\Delta t = 24$ hours
  - `LOW`: $\Delta t = 48$ hours
- **FR-5.3 [Automated DAA Escalation Engine]:** The system shall evaluate pending grievances against the current timestamp. Any grievance where $\text{Current Time} > \text{Deadline}$ with status `OPEN` shall transition to `ESCALATED`, setting `is_escalated = True` and notifying the Dean Academic Affairs dashboard.
- **FR-5.4 [Grievance Resolution]:** The concerned department in-charge or DAA may resolve a grievance with resolution notes, optionally updating or waiving the associated dues.

Module 6: Certificate Generation & Public Verification

- **FR-6.1 [Cryptographic Hash Calculation]:** The certificate hash shall be computed as: \text{Hash}_{\text{cert}} = \text{SHA-256}\Big(\text{cert_id} \parallel \text{student_sid} \parallel \text{academic_year} \parallel \text{completion_timestamp}\Big)
- **FR-6.2 [Official PDF Certificate Rendering]:** The system shall dynamically generate a print-ready vector PDF document bearing:

    - Official Punjab Engineering College header, crest, and Sector 12 address.
    - Student details (Full Name, 8-digit SID, Academic Program, Branch).
    - Departmental clearance breakdown with timestamps and positional titles.
    - Caution Money refund confirmation in INR (`₹`).
    - Cryptographic verification SHA-256 hash.
    - Signature blocks for Dean Academic Affairs (DAA) and Dean Student Affairs (DSA).
- **FR-6.3 [Public Hash Verification]:** The system shall provide a public REST endpoint `/api/v1/certificate/verify` accepting `certificate_id` or `certificate_hash`. It shall return verification status, student name, SID, issuance date, and validity confirmation without requiring user authentication.

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4. Non-Functional Requirements (NFR)

4.1 Performance Requirements

- **NFR-1.1 [API Latency]:** Under normal operating load (100 concurrent requests), 95% of read queries shall respond within $\leq 150\ \text{ms}$, and 95% of state modification transactions shall complete within $\leq 300\ \text{ms}$.
- **NFR-1.2 [Certificate Rendering Throughput]:** Dynamic vector PDF generation shall complete in $\leq 1.5\ \text{seconds}$.
- **NFR-1.3 [SLA Evaluation Execution]:** The automated SLA escalation scan shall complete evaluation across all active tickets in $\leq 500\ \text{ms}$.

4.2 Security & Integrity Requirements

- **NFR-2.1 [Token-Based Authentication]:** API communication shall be authenticated via cryptographically signed JWT bearer tokens with expiration limits.
- **NFR-2.2 [Cryptographic Signatures]:** Departmental seals must use HMAC-SHA256 with a private institutional server key. Any tampering with stored approval records shall invalidate the digital signature.
- **NFR-2.3 [Input Sanitization & Injection Prevention]:** All database operations must be executed through parameterized ORM statements to eliminate SQL injection vulnerabilities.

4.3 Concurrency & Idempotency Requirements

- **NFR-3.1 [Concurrent Approval Consistency]:** Concurrent approvals by Gate 1 departments (Hostel, Library, Sports) shall execute idempotently without deadlocks or lost updates.
- **NFR-3.2 [Client-Side Synchronization]:** Client dashboards shall synchronize with the server state within an interval of $\leq 10\ \text{seconds}$ to reflect cross-user updates.

4.4 Reliability & Maintainability

- **NFR-4.1 [Fault Isolation]:** Failure in one departmental workflow (e.g., temporary unavailability of Sports staff) shall not block independent operations in other Gate 1 departments.
- **NFR-4.2 [Implementation Independence]:** Architectural contracts between frontend clients, workflow handlers, and data stores shall be decoupled via RESTful interfaces, allowing database engine swapping (SQLite $\rightarrow$ PostgreSQL) with zero changes to business logic.

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5. Architectural & System Boundaries (Frozen Invariants)

The following architectural invariants are **strictly frozen** and shall remain unchanged across any code refactoring or feature additions:

1. **Three-Gate Structural Topology:** The 3-gate progression ($\text{Gate 1 [Parallel]} \rightarrow \text{Gate 2 [Labs]} \rightarrow \text{Gate 3 [Accounts]}$) is invariant.
2. **Positional Staff Designations:** System roles shall remain role-based and positional (DAA, Chief Warden, Librarian, Gymkhana President, HOD Labs, Accounts Officer).
3. **Monetary Unit of Account:** The system's base currency is Indian Rupees (`₹` / INR).
4. **Institutional Identifier:** Student identification is bound to the 8-digit PEC SID format (`YYBBBRRR`).
5. **Cryptographic Validation Contract:** Certificate verification relies on SHA-256 digest comparison against immutable stored approval records.